

A Research Programme in the Modelling, Simulation and Control of the Electric Arc Furnace (EAF) Steelmaking Process

*Critical Literature Survey, Research Methodology, and a Proposed Joint Master's /
Doctoral Programme*

Toward a validated, calibratable digital-twin and decision-support platform for industrial EAF
operation, process study and improvement

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Executive Summary

Electric arc furnace (EAF) steelmaking is the central technology of the scrap- and direct-reduced-iron (DRI)-based, lower-carbon route to steel. It is, however, one of the most energy-intensive and operationally complex unit processes in the metallurgical industries: strongly coupled electrical, thermal and chemical phenomena unfold together in an aggressive, poorly observable environment, and small changes in charge mix, power profile, oxygen and carbon practice or timing translate into large differences in energy consumption, metallic yield, productivity and emissions. Mathematical modelling and simulation are therefore indispensable both for understanding the process and for guiding its operation and improvement.

This report critically reviews a corpus of twenty-seven scholarly and technical works spanning the principal families of EAF models — comprehensive reviews; static mass-energy balances and energy-efficiency analyses; dynamic lumped-zone process models; rigorous thermochemical and slag-metal reaction models; electrical and arc models; computational-fluid-dynamics (CFD) studies of scrap and DRI melting; and DRI/hydrogen-DRI process models — together with their applications to decision support, online optimisation and operator training. From this synthesis the report identifies a coherent set of research gaps and, on that basis, proposes a joint Master's/Doctoral programme to build deep, transferable expertise in the EAF process and to deliver a validated, calibratable digital-twin and decision-support platform that is directly useful to industry for studying their processes and recommending modifications.

The proposed programme is organised as a physics-plus-data research strategy delivered through seven work packages, phased so that a Master's project establishes a validated static-plus-dynamic model, sensitivity and calibration methodology and an operator decision-support tool on a partner plant's data, while a subsequent Doctoral project advances hybrid physics-informed machine-learning surrogates and soft sensors, rigorous but real-time thermochemistry, extension to DRI/H₂-DRI and green-steel scenarios, and closed-loop optimisation and control. A working static-plus-dynamic EAF model with an operator graphical interface and a sensitivity-analysis suite — developed as a precursor to this proposal — provides a concrete software foundation on which the programme can build immediately (Appendix B).

1 Introduction and Background

1.1 The electric arc furnace in modern steelmaking

The EAF melts a metallic charge — principally steel scrap, increasingly supplemented or replaced by DRI/HBI and, in some shops, hot metal — using the intense heat of high-power electric arcs struck between graphite electrodes and the charge, augmented by chemical energy from oxy-fuel burners, oxygen lancing and carbon injection. A heat proceeds through charging, bore-in, melt-down, and a flat-bath refining period in which the bath is decarburised, dephosphorised, heated to tap temperature and covered by a foaming slag, before tapping into a ladle for secondary metallurgy and casting. Compared with the integrated blast-furnace/basic-oxygen route, the EAF route has a far smaller direct CO₂ footprint, is flexible in scale and charge, and is the natural vehicle for recycling steel and for consuming DRI produced from natural gas or, prospectively, hydrogen.

1.2 Global and Indian context; the decarbonisation driver

EAF- and induction-based electric steelmaking already accounts for a large and growing share of world crude-steel output, and its share is expected to rise sharply as the industry decarbonises, as scrap availability grows and as DRI capacity expands. This trajectory is especially pronounced in India, where a very large fraction of steel is made through electric routes fed by coal- and gas-based sponge iron (DRI) together with scrap. India's combination of DRI-heavy charges, variable scrap and raw-material quality, and strong pressure to reduce specific energy consumption and emissions makes rigorous, plant-calibrated EAF process modelling both timely and of direct national industrial relevance. The emergence of hydrogen-DRI and the drive toward near-carbon-neutral steel add further urgency to understanding melting and refining behaviour under high-DRI and mixed charges.

1.3 Why model the EAF?

Three features make the EAF an outstanding candidate for modelling. First, it is energy-intensive: even a few percent of specific-energy improvement, spread across millions of tonnes, represents a very large economic and environmental gain. Second, it is complex and tightly coupled: electrical supply and arc behaviour, heat transfer and scrap melting, slag-metal-gas thermochemistry, foaming and off-gas dynamics all interact, so intuition alone is an unreliable guide. Third, it is poorly observable: the interior is hostile to instrumentation, measurements (bath temperature, carbon) are intermittent and late, and much of the operator's knowledge is tacit. Models turn sparse measurements and first-principles understanding into a continuous, quantitative picture of the heat, enabling what-if analysis, set-point optimisation, soft sensing, operator training and, ultimately, closed-loop control.

1.4 Purpose and structure of this report

This report has two purposes: to provide a critical survey of the EAF-modelling literature represented by the supplied corpus, and to translate that survey into a concrete, defensible research programme. Section 2 reviews and critically analyses the literature by theme and distils the research gaps. Section 3 states the problem, research questions and objectives. Section 4 sets out the research methodology and approach in detail. Section 5 specifies the proposed joint Master's/Doctoral programme, its work packages, phasing and timeline. Section 6 discusses expected outcomes, novelty and industrial impact; Section 7 covers resources, collaboration and risk; Section 8 concludes. Appendix A tabulates the reviewed corpus; Appendix B describes the software foundation already in place.

2 Critical Literature Survey

2.1 Method and scope of the survey

The corpus comprises twenty-seven items (a small number are duplicate copies, and one is an image-only slide deck), ranging from peer-reviewed journal articles and a research monograph to a doctoral dissertation and conference/industrial presentations. They were read and classified by modelling philosophy, the phenomena they resolve, their intended use, and their validation basis. The survey is organised thematically rather than chronologically so that competing approaches to the same problem can be compared directly. Each theme is summarised, then critically appraised for strengths, limitations and open questions; Section 2.12 synthesises across themes and Section 2.13 sets out the resulting research gaps. A structured, per-paper summary is provided in Appendix A.

2.2 A taxonomy of EAF models

EAF models can be arranged along a spectrum of spatial and physical resolution, from zero-dimensional balances to fully three-dimensional multiphysics, with a growing branch of data-driven and hybrid approaches. The trade-off along this spectrum — between physical fidelity and the speed, robustness and calibratability needed for online use — is the single most important organising theme of the literature and of this proposal.

Model family	Representative works in corpus	What it resolves	Typical use
Reviews / overviews	Abadi et al. (2024); Hay et al. (2020); Conejo (2024, monograph)	Landscape, taxonomy, energy balance, methods	Orientation; identifying gaps
Static mass & energy balance	Energy-efficiency analysis; Conejo (2024)	Per-heat charge, flux, oxygen and energy requirements; yield	Heat planning; benchmarking
Dynamic lumped / zone models	Logar et al. Part 1 & 2 (2012); MacRosty & Swartz (2005); Bekker et al. (1999); Opitz & Treffinger (2016); Hay/Meier lineage; DynEAF	Time evolution of temperature, chemistry, slag, energy in a few zones	Online estimation; optimisation; training
Rigorous thermochemistry	Van Ende EERZ (2022); Hay et al. slag/metal chemistry (2019)	Slag–metal–gas equilibria & reaction kinetics with databases	Endpoint chemistry; refining
Electrical / arc models	Lozynskyy et al. (Energies)	Arc electrics, impedance, electrode regulation, grid impact	Electrode control; power quality
CFD / multiphysics melting	Chen et al. CFD (scrap); Chen thesis (2022); González et al. (2010); Pineda-Martínez et al. (2015)	3-D fields; particle/scrap melting kinetics; frozen shell	Design; deep phenomenology
DRI / H2-DRI process models	Morales/Conejo (2001, 2004); Wang/Li mixed charge	Melting & refining with DRI, FeO reduction, foaming	DRI practice; green-steel charges
Decision support / control	Decision-support system; AIST online balance; simulators	Model wrapped for monitoring, advice, optimisation	Operator guidance; closed loop

2.3 Review and overview literature

Breadth versus depth. The review by Makki Abadi, Tang and Rashidi (Heliyon, 2024) surveys simulation and numerical modelling of the EAF and its sub-processes, covering CFD, energy balance and optimisation, and situating the technology within global steel production. It is valuable as a map of methods and as evidence of the field's breadth, but, like most single-article reviews, it necessarily treats each method briefly and does not reconcile the competing modelling philosophies quantitatively.

A modelling-focused review. Hay, Visuri, Aula and Echterhof (steel research international, 2020) review mathematical process models specifically, tracing the lineage of lumped dynamic models and thermochemical treatments. This is the most directly relevant review for the present programme because it foregrounds the process-model family from which most control-oriented work descends, and it makes explicit the recurring tension between chemical rigour and computational tractability.

A practitioner's synthesis on energy. Conejo's monograph, *Electric Arc Furnace: Methods to Decrease Energy Consumption* (Springer, 2024), consolidates the energy engineering of the EAF — electrical and chemical energy, losses, foaming, scrap preheating, and best-practice levers for reducing kWh/t. It is an authoritative reference for the energy-balance components used throughout the corpus and for framing the industrial objective of the proposed work.

Critical appraisal. Collectively the reviews establish the taxonomy and the central fidelity/tractability trade-off, but they also reveal what the primary literature under-serves: systematic cross-plant validation, uncertainty quantification, and the integration of data-driven methods with first-principles models. These absences motivate the programme's emphasis on calibration, validation and hybrid modelling.

2.4 Static mass-energy balance and energy-efficiency models

Static models compute, for a whole heat, the charge and flux requirements, the oxygen and carbon demand, the slag amount and basicity, the metallic yield and — through an energy balance of sinks (heating and melting steel and slag, off-gas and shell losses) against sources (electrical input, oxidation chemical energy, post-combustion, burners and any hot-metal sensible heat) — the electrical energy required. The energy-efficiency analysis in the corpus exemplifies this class, decomposing the energy streams and quantifying where energy is lost; Conejo's monograph provides the underlying engineering.

Strengths. Static balances are transparent, fast and robust; they are the natural tool for heat planning, benchmarking and first-order what-if studies, and they require only modest data.

Limitations. By construction they carry no time information: they cannot represent melt-down dynamics, the carbon boil, foaming evolution, or the trajectory of temperature and chemistry, and they treat oxidation and yield through fixed fractions rather than mechanistically. They are therefore a necessary foundation but an insufficient basis for control or for diagnosing dynamic process problems — which is precisely why the dynamic models of Section 2.5 exist.

2.5 Dynamic lumped-zone process models

This family is the backbone of control-oriented EAF modelling. The furnace interior is divided into a small number of interacting zones — typically solid scrap, liquid metal, solid and liquid slag, and gas, plus roof and wall — each with an energy (and mass) balance, and the resulting system of ordinary differential equations is integrated through the heat. The models are fast enough for online use yet retain the essential physics.

Logar, Dovzan and Skrjanc (ISIJ, 2012, Parts 1 and 2). This two-part work is the most influential control-oriented formulation in the corpus and is explicitly aimed at energy-consumption optimisation and operator-training simulators. Part 1 develops the heat- and mass-transfer structure — the zone energy balances, a physically elegant melting formulation in which the energy delivered to solid scrap is split continuously between heating and melting, mass re-averaging of temperature on charging, radiative exchange via view factors, and burner and post-combustion terms. Part 2 supplies the thermochemistry — decarburisation and the oxidation of Si, Mn, Cr and P, FeO formation and reduction, and slag chemistry — expressed through equilibrium relations and reaction rates. The model is validated against industrial heats. Its enduring value is the balance it strikes between fidelity

and tractability; its principal simplifications are the geometric radiation network (which needs many view-factor/area coefficients) and the equilibrium-constant chemistry (which trades rigour for speed).

MacRosty and Swartz (Ind. Eng. Chem. Res., 2005). This model represents the reacting system as equilibrium zones with mass-transport limitations, deliberately engineered for continuity and robustness so that it can be embedded in an optimisation framework, with parameters estimated from industrial data. It is an important demonstration that a dynamic EAF model can be made optimisation-ready, and it anticipates the model-based control theme; its equilibrium-zone abstraction, however, still requires careful parameter estimation and does not resolve spatial detail.

Bekker, Craig and Pistorius (ISIJ, 1999). An early and much-cited multivariable, non-linear state-space model built specifically to test control strategies using off-gas variables, with physical states such as bath carbon and rates such as decarburisation. It established the state-space, control-theoretic view of the EAF and the use of off-gas as an information source; being of its time, it is comparatively coarse chemically and thermally.

Opitz and Treffinger (Metall. Mater. Trans. B, 2016). A dynamic, physics-based model of the complete plant — vessel, electric system, electrode regulation and off-gas system — implemented in the equation-based language Modelica, with a simple spatial discretisation of the solid so that arc and burner energy input depend on melting progress. Its distinctive contribution is the plant-level, multi-subsystem coupling and the object-oriented implementation; the trade-off is added complexity and a still-simplified chemistry.

The Meier/Hay lineage and DynEAF. A sustained line of work at RWTH Aachen (Hay, Echterhof, Pfeifer and colleagues) has developed and progressively enriched a comprehensive dynamic process model: Hay, Echterhof and Visuri built a simulator with real-time user input and an algorithm for automatically generating operation charts, explicitly for operator training and for rapid evaluation of many scenarios; the DynEAF development reports (Echterhof, Hay, Pfeifer) add variable melting geometry, improved thermal-radiation and gas-phase modelling and adaptable carbon sources. This lineage is the closest in spirit to a deployable engineering tool and to the present proposal; its models are correspondingly data-hungry and rely on numerous empirical sub-models that must be calibrated.

Critical appraisal of the family. Lumped-zone models are the right level of abstraction for online use, but three weaknesses recur across all of them: (i) they depend on many empirical coefficients (heat-transfer, exposure, foaming, kinetic rates) that are plant-specific and are usually fitted to a single furnace, so transferability is unproven; (ii) radiation and melting geometry are represented approximately; and (iii) chemistry is simplified relative to database thermodynamics. These are exactly the weaknesses that rigorous thermochemistry (Section 2.6), CFD phenomenology (Section 2.8) and data-driven calibration (Section 2.13) can address, and that this programme sets out to address in an integrated way.

2.6 Thermochemical and slag-metal reaction modelling

The EERZ approach — Van Ende (JOM, 2022). The Effective Equilibrium Reaction Zone model couples kinetic expressions with accurate thermodynamic databases to predict the evolving mass, temperature and composition of scrap, liquid metal, slag and gas, representing charge heating, scrap melting, C and O₂ injection, slag-metal-gas reactions and post-combustion through empirical links to process conditions. It reproduces measured endpoint temperature and slag/metal composition on industrial furnaces and illustrates the benefit of the hot-heel practice. Its strength is chemical rigour grounded in computational thermodynamics; its cost is heavier computation and dependence on a thermodynamic database and effective reaction-zone/volume parameters.

Improved chemistry within a dynamic model — Hay, Reimann and Echterhof (Metall. Mater. Trans. B, 2019). Building directly on the Logar/Meier model, this work upgrades the thermochemical module:

it introduces separate reaction zones for slag-metal interaction and for oxygen and carbon injection, computes equilibrium compositions and reaction rates per zone, treats diffusion as a rate-limiting step from the bulk melt to the reaction zone, adds previously neglected species (dissolved O and S, CaS), and simplifies the carbon treatment for accuracy and speed; it is validated against extensive industrial data and quantifies the effect of different activity models on both accuracy and run time. This paper is a model of how to increase chemical rigour inside a tractable dynamic framework — and it makes explicit the accuracy-versus-runtime trade-off that any online deployment must manage.

Critical appraisal. The thermochemical strand shows that database-grade slag-metal chemistry can be embedded in EAF models, materially improving endpoint prediction. The open problem is doing so at online speed and with parameters that transfer across plants; coupling reduced-order surrogates of the thermodynamics with dynamic models is a clear opportunity.

2.7 Electrical operation, arc modelling and electrode control

Lozynskyy et al. (Energies). This work synthesises a real-time model of the electrical mode of the arc furnace, suitable for analysing the furnace's impact on the supply network at different melting stages and control systems, and for building model-predictive control of electrode movement, including hardware-in-the-loop. It represents the electrical/arc sub-system that most process models treat only through an efficiency or a prescribed power profile.

Critical appraisal. The electrical and process literatures are largely disjoint: electrical/arc models resolve impedance, flicker and electrode dynamics but abstract the metallurgy, while process models abstract the electrics. Yet arc stability, power input and electrode regulation directly shape melting and energy transfer (and are strongly affected by foaming). Integrated electrical-thermal-chemical modelling — and control that respects both — is under-developed and is an explicit target of this programme.

2.8 Scrap and DRI melting kinetics: fundamental and CFD studies

Particle and pellet melting — González, Ramírez-Argáez and Conejo (ISIJ, 2010); Pineda-Martínez et al. (ISIJ, 2015). The first couples a CFD field to a Lagrangian model of melting/solidifying DRI particles, quantifying how initial particle size and temperature, feed position and rate, and arc length govern melting rate, and resolving the frozen shell that forms and re-melts in the early stage. The second models the melting of an isolated sponge-iron pellet in non-reactive slag, and identifies the convective heat-transfer coefficient between bath and particle as the controlling parameter for both melting time and slag-shell thickness. Together they provide the mechanistic basis for the melting sub-models that lumped models represent with a single coefficient.

Scrap-pile melting by CFD — Chen et al.; Chen (PhD, Purdue, 2022). This work develops a three-dimensional integrated CFD model of scrap melting in an AC EAF using dual-cell and stack approaches for the dynamic melting and collapse of the scrap pile, integrated with an electric-arc model and a coherent-jet (oxygen) model, a Monte-Carlo treatment of arc radiation and an electrode-regulation strategy — and, importantly, validates it against purpose-designed experiments on an industry-scale 150-tonne furnace. It is among the most complete phenomenological treatments of melt-down in the corpus.

Critical appraisal. CFD and Lagrangian studies deliver the deepest physical insight and the mechanistic closures (heat-transfer coefficients, melting-rate laws, frozen-shell behaviour) that lumped models need — but they are far too computationally expensive for online use or for scanning many heats. This complementarity is central to the proposed strategy: use high-fidelity simulation and experiment to derive and calibrate the reduced-order closures that the dynamic, deployable model uses.

2.9 DRI- and hydrogen-DRI-based EAF process models

The Morales–Conejo simulators (ISIJ, 2001; Metall. Mater. Trans. B, 2004). The 2001 simulator models the melting of DRI in the EAF, tracking the time evolution of temperature and metal/slag chemistry from cost-optimised mass and energy balances that set the initial and boundary conditions, and controlling bath oxidation through carbon-fines injection into the slag; a key finding is that the final bath-oxidation level depends more on the process route (feeding sequence, carbon and oxygen practice, slag basicity, carbon wettability) than on DRI quality itself. The 2004 companion advances the phenomenology of FeO reduction (by dissolved carbon and by injected carbon particles, including surfactant effects) and introduces a Dynamic Foaming Index to quantify real slag foaming through a heat. These works are foundational for DRI practice and for foaming.

Mixed DRI + scrap and hydrogen-DRI — Li, Wang et al. This recent phenomenological model integrates materials-balance and thermochemistry modules for an EAF charged with a mixture of hydrogen-DRI and scrap, covering scrap and DRI melting, slag formation, carbon and oxygen injection and heterogeneous reactions, and predicting slag and steel composition in real time. It quantifies how the DRI share and DRI chemistry (metallisation, carbon) affect process dynamics, and concludes that a well-calibrated carbon/oxygen strategy matched to DRI metallisation and carbon is needed to maximise yield while limiting bath oxidation. It directly addresses the green-steel charge that is strategically important for India.

Critical appraisal. The DRI strand is metallurgically mature but is dominated by a few groups and a few plants; models for high-DRI and H₂-DRI charges, continuous feeding, gangue and energy penalties, and their interaction with foaming and yield remain comparatively sparse and under-validated across the diversity of DRI qualities (notably coal-based sponge iron prevalent in India). This is a high-value, high-relevance gap.

2.10 Foaming-slag modelling

Foaming slag is decisive for EAF energy efficiency: a well-foamed slag shields the arc, raising energy transfer to the bath and protecting the refractories and water panels, while poor foam wastes energy to the panels and lengthens the heat. The corpus treats foaming through the Dynamic Foaming Index of Morales–Conejo, which links measurable foaming behaviour to CO generation and slag properties, and through the foaming and FeO-reduction terms embedded in the Logar and Hay models. Critically, foaming remains among the most empirical parts of every model: it depends on CO evolution rate, slag viscosity and surface properties, basicity and FeO, in ways that are still represented by correlations rather than predicted from first principles. Predictive, chemistry-linked foaming — and its use in arc-coverage-aware control — is an open problem of high practical value.

2.11 Decision support, online optimisation and operator training

Online balance and optimisation. The AIST dynamic energy-and-material-balance model is designed for on-line process optimisation, tracking the heat's energy and mass state in real time to support set-point decisions. It exemplifies the operational deployment of a balance/process model.

Model-based decision support. The model-based decision-support system for EAF online monitoring and control wraps a process model as a real-time monitor that estimates the heat state and advises the operator on set-point changes to reduce energy and stabilise the process. This is the paradigm the present programme targets: a model that not only simulates but guides.

Simulators for training. Several works above — the Hay/Echterhof simulator with automatic operation charts, and the explicit operator-training aim of Logar et al. — show that dynamic models double as training and scenario-evaluation tools, letting operators and engineers explore the consequences of changes safely and quickly.

Critical appraisal. Decision-support and online applications are demonstrated but remain comparatively few and are typically single-plant. Robust state estimation from sparse measurements, uncertainty-aware advice, and the step from open-loop advice to trustworthy closed-loop control are the frontier — and require the calibration, soft-sensing and control research proposed here.

2.12 Critical synthesis across the literature

Read together, the corpus tells a consistent story. There is a well-developed continuum of models from static balances through lumped-zone dynamic models to full CFD/multiphysics, and a parallel strand of increasingly rigorous slag-metal thermochemistry. The lumped-zone dynamic model — exemplified by Logar et al. and the RWTH lineage, made optimisation-ready by MacRosty and Swartz and control-oriented by Bekker et al. — is the established sweet spot for online use, and it has been demonstrated for decision support, optimisation and operator training. High-fidelity CFD and thermodynamic-database models supply the mechanistic closures and endpoint chemistry that the lumped models approximate. DRI and, now, hydrogen-DRI charges are an active and strategically important frontier.

Equally consistent, however, are the limitations. Nearly every dynamic model depends on numerous empirical, plant-specific coefficients that are fitted to a single furnace, with little systematic treatment of calibration methodology, uncertainty, or transfer across plants. Chemistry and foaming are simplified for speed. The electrical/arc sub-system and the metallurgical process are usually modelled in isolation. Data-driven and hybrid physics-informed machine-learning methods — soft sensors, surrogates, digital twins, learning-based control — are strikingly under-represented in this corpus, even though the EAF's poor observability and rich data make it an ideal application. And validation, while present, is rarely cross-plant or uncertainty-quantified. These are not criticisms of individual excellent papers; they are the field-level gaps that a focused research programme can convert into contributions.

2.13 Research gaps and opportunities

The following gaps emerge directly from the critical synthesis and frame the proposed programme.

#	Gap identified in the literature	Evidence from the corpus	Opportunity / proposed contribution
G1	Fidelity vs. real-time tractability	CFD/multiphysics (Chen; González) accurate but slow; lumped models (Logar; Hay) fast but approximate	Reduced-order and surrogate models that inherit CFD-grade closures; hybrid physics-ML
G2	Calibration, uncertainty & cross-plant transfer	Most models fitted to one furnace; little UQ or transfer methodology	Formal parameter estimation, Bayesian calibration, UQ and transfer-learning across plants
G3	Rigorous chemistry at online speed	EERZ (Van Ende) and Hay et al. are rigorous but heavy; Logar simplifies	Fast thermodynamic surrogates coupled to the dynamic model
G4	Data-driven & hybrid modelling	Corpus is almost entirely first-principles; ML/soft sensors scarce	Physics-informed ML, soft sensors for T/C/FeO, digital twin, data assimilation
G5	DRI / H2-DRI & green-steel charges	Emerging (Wang/Li; Morales/Conejo); sparse for high-DRI, coal-DRI, continuous feed	Validated models for high-DRI/H2-DRI and Indian sponge-iron practice
G6	Electrical-thermal-chemical coupling	Electrical (Lozynskyy) and process models are disjoint	Integrated arc-thermal-chemical model and coupling-aware control
G7	Predictive foaming & arc coverage	Foaming still empirical (DFI; correlations)	Chemistry-linked foaming model; arc-coverage-aware energy transfer
G8	From advice to closed-loop control	Decision support and MPC demonstrated but rarely closed-loop	Robust state estimation + NMPC / reinforcement-learning control, safely deployed

3 Problem Statement, Research Questions and Objectives

3.1 Problem statement

Despite four decades of EAF modelling, the industry still lacks a single, validated, calibratable modelling-and-decision-support platform that is simultaneously (i) fast and robust enough for online use and operator guidance, (ii) chemically and physically faithful enough to be trusted for process study and modification, (iii) systematically calibrated to a specific plant with quantified uncertainty and demonstrated transfer to others, and (iv) capable of representing modern high-DRI/hydrogen-DRI charges. The consequence is that many shops still operate on heuristics and lagging measurements, leaving recoverable energy, yield, productivity and emissions on the table.

3.2 Research questions

RQ1. How can a reduced-order dynamic EAF model retain the accuracy of high-fidelity (CFD/thermodynamic-database) models while remaining fast and robust enough for online estimation, optimisation and operator guidance?

RQ2. What calibration, uncertainty-quantification and transfer methodology allows such a model to be tuned to a specific furnace from routine plant data, with quantified confidence, and adapted to other furnaces with minimal re-work?

RQ3. Can hybrid physics-informed machine-learning soft sensors provide reliable real-time estimates of otherwise unmeasured or lagging quantities (bath temperature, carbon, slag FeO), and improve endpoint prediction over first-principles models alone?

RQ4. How do high-DRI and hydrogen-DRI mixed charges alter melting, refining, foaming, yield and energy, and how should carbon/oxygen/power practice be adapted — particularly for Indian sponge-iron qualities?

RQ5. Can a coupled electrical-thermal-chemical model and a model-based control layer (set-point optimisation, NMPC and/or reinforcement learning) deliver measurable, safe improvements in energy, yield and productivity in an industrial setting?

3.3 Aim and objectives

Aim. To develop, validate and industrially demonstrate a calibratable, hybrid physics-plus-data EAF modelling and decision-support platform — spanning static and dynamic models, soft sensing and control — that builds deep expertise in the process and is directly useful to industry for studying their operations and recommending modifications.

The aim is pursued through six objectives:

- O1.** Consolidate and extend a modular static-plus-dynamic EAF model (heat/mass transfer, melting, thermochemistry, electrical coupling, off-gas and foaming), building on the established lumped-zone lineage and the software foundation in Appendix B.
- O2.** Acquire, curate and analyse industrial heat data from a partner plant; develop and apply a formal calibration, sensitivity-analysis and uncertainty-quantification methodology, and test cross-plant transfer.
- O3.** Develop hybrid physics-informed machine-learning surrogates and soft sensors for real-time estimation and reduced-order online deployment, and quantify their accuracy against plant data.
- O4.** Extend and validate the model for DRI and hydrogen-DRI mixed charges relevant to Indian and global green-steel practice.

- O5.** Develop a process-optimisation and advanced-control layer (multi-objective set-point optimisation, non-linear model-predictive and/or reinforcement-learning control) and an operator decision-support/digital-twin deployment.
- O6.** Conduct industrial case studies that quantify energy, yield, productivity and emission improvements and translate them into concrete process-modification recommendations.

4 Research Methodology and Approach

4.1 Overall research strategy

The programme follows a physics-plus-data strategy that deliberately exploits the complementarity identified in the survey: first-principles models provide structure, interpretability and extrapolation; high-fidelity simulation and experiment provide mechanistic closures; and plant data provide calibration, soft sensing and validation. The strategy is delivered in five methodological threads — model development; data and calibration; hybrid/data-driven modelling; process extension (DRI/H2-DRI); and optimisation, control and deployment — woven through the work packages of Section 5. Each modelling claim is tied to a validation test against independent industrial heats, and every parameter that materially affects predictions is subjected to sensitivity analysis and, where possible, uncertainty quantification.

4.2 Model development

Architecture. A modular architecture separates the static balance, the dynamic zone model, the thermochemistry, the melting closures, the electrical coupling, the off-gas/foaming sub-models and the diagnostics/guidance layer, so that each can be independently improved, replaced by a higher-fidelity or surrogate variant, and calibrated. This modularity is already realised in the software foundation (Appendix B).

- **Static model.** Elemental mass balance and an itemised energy balance (heating/melting of steel and slag, off-gas and shell losses versus electrical, oxidation, post-combustion, burner and hot-metal energy), yielding tap weight, yield, flux/oxygen/carbon demand, slag chemistry and specific energy — used for planning, benchmarking and as boundary/initial conditions for the dynamic model (following the Morales-Conejo and energy-efficiency practice).
- **Dynamic zone model.** Coupled ODEs for the solid-scrap, liquid-metal, slag and gas zones — energy balances with the continuous heating/melting split and mass re-averaging of Logar et al., an immersion-melting closure, and a refining-chemistry set (decarburisation with an oxygen-limited/mass-transfer-limited regime change, Si/Mn/P oxidation, FeO formation and carbon reduction), CO post-combustion and a foaming index that modulates arc-transfer efficiency.
- **Thermochemistry upgrade path.** Starting from equilibrium-constant/first-order-rate chemistry and providing a path to database-grade thermochemistry via the EERZ concept (Van Ende) and the zone/diffusion-limited treatment of Hay et al., including dissolved O and S and CaS, with fast thermodynamic surrogates (Section 4.8) for online speed.
- **Electrical coupling.** From an efficiency/power-profile representation toward an integrated arc/electrical sub-model (after Lozynskyy et al.) so that arc power, electrode regulation and foaming-dependent transfer are represented consistently (addresses G6).
- **Melting closures.** Heat-transfer and melting-rate closures for scrap and DRI/pellets informed and calibrated by the CFD/Lagrangian studies (Chen; González; Pineda-Martínez), rather than by single fitted coefficients.

4.3 Numerical methods and software engineering

The dynamic system is integrated with robust fixed- or adaptive-step schemes chosen for stability and speed; continuity and boundedness of all sub-model expressions are enforced so that the model is optimisation- and control-ready (a lesson from MacRosty and Swartz). The platform is engineered as maintainable, version-controlled, tested software with a clear API, a parameter registry with documentation and validation, a sensitivity/uncertainty toolkit, and an operator graphical interface —

so that it is simultaneously a research instrument and an industrially deployable digital twin. The existing implementation (Appendix B) provides this scaffolding.

4.4 Industrial data acquisition and management

A memorandum with one or more partner steel plants will provide historical and, later, live heat data: level-1/level-2 and SCADA/PLC signals (active power and electrical parameters, electrode positions, oxygen and carbon injection, burner firing, additions, timings), off-gas analysis and flow where available, charge descriptions and scrap/DRI analyses, and laboratory tap analyses and temperatures. Data will be cleaned, time-aligned, quality-flagged and organised into a heat database, with careful treatment of missing and lagging measurements. Where controlled trials are permissible, a design-of-experiments plan will vary selected set-points to enrich the identifiability of key parameters.

4.5 Calibration, parameter estimation and uncertainty quantification

Addressing G2, model parameters (heat-transfer and melting coefficients, kinetic rates, foaming and efficiency parameters) will be estimated from plant data by non-linear least squares and Bayesian inference, distinguishing parameters that are physically transferable from those that are plant-specific. Uncertainty will be propagated to predictions (Monte-Carlo and, where appropriate, polynomial-chaos or Gaussian-process emulation), so that the model reports confidence, not just point values. Cross-plant transfer will be tested by re-calibrating a minimal parameter subset on a second furnace, quantifying the effort and accuracy of transfer.

4.6 Sensitivity analysis

Both local (one-factor-at-a-time, tornado) and global (variance-based Sobol, Morris screening) sensitivity analyses will identify the parameters and operating levers that most influence energy, yield, temperature, chemistry and productivity — guiding calibration effort, model reduction and operating recommendations. A sensitivity suite producing multi-panel static and dynamic sweeps, response-surface heatmaps and tornado rankings already exists (Appendix B) and will be extended with global methods.

4.7 Validation strategy

Validation is layered: (i) component validation of sub-models against the fundamental studies and laboratory/CFD results (e.g. melting-rate laws, thermodynamic endpoints); (ii) whole-heat validation against held-out industrial heats not used in calibration, comparing predicted and measured tap temperature, carbon, slag chemistry, energy and tap-to-tap time; (iii) trajectory validation of the time evolution where intermittent measurements allow; and (iv) cross-plant validation. Acceptance criteria (e.g. error bands on tap temperature, carbon, kWh/t) will be agreed with the industrial partner.

4.8 Hybrid physics-informed and data-driven modelling

Addressing G1, G3 and G4 — the most significant gap in the corpus — the programme will develop hybrid models that combine first-principles structure with machine learning:

- **Soft sensors.** Data-driven estimators (Gaussian processes, gradient-boosted trees, recurrent/temporal neural networks) for real-time inference of bath temperature, carbon and slag FeO from routinely measured signals, trained and tested against laboratory analyses.
- **Thermodynamic surrogates.** Fast surrogates of database equilibrium calculations, enabling EERZ-grade chemistry at online speed within the dynamic model.
- **Gray-box and physics-informed learning.** Residual/gray-box models and physics-informed neural networks that learn the discrepancy between the physical model and plant data while

respecting conservation laws, improving accuracy without sacrificing interpretability or extrapolation.

- **Data assimilation.** State-estimation (extended/ensemble Kalman filtering, moving-horizon estimation) that fuses the dynamic model with streaming measurements to track the heat in real time and underpin control.

4.9 Extension to DRI / hydrogen-DRI and green-steel scenarios

Addressing G5, the melting and refining sub-models will be extended and validated for DRI and hydrogen-DRI mixed charges — metallisation and residual FeO, gangue and its energy/flux penalty, carbon content, continuous feeding, and the reduction of FeO by dissolved and injected carbon with foaming — building on Morales-Conejo and Wang/Li and calibrated to the sponge-iron qualities relevant to Indian practice. Scenario studies will map energy, yield, foaming and bath-oxidation trade-offs across DRI share and quality, and derive recommended carbon/oxygen/power practices.

4.10 Process optimisation and advanced control

Addressing G6 and G8, the calibrated, control-ready model will support: multi-objective optimisation of the operating strategy (energy, yield, productivity, emissions) using the sensitivity structure to reduce the search space; non-linear model-predictive control (NMPC) built on the dynamic model and the data-assimilation estimator; and, as a research thread, reinforcement-learning control trained in the simulator and constrained for safe operation. Electrode-regulation/electrical control will be integrated with process control so that arc stability, foaming and energy transfer are managed together.

4.11 Operator decision-support and digital-twin deployment

Following the decision-support and simulator paradigm (AIST online balance; the decision-support system; the Hay/Echterhof training simulator; Logar's training aim), the platform will be deployed as an operator-facing digital twin: real-time state estimation and prediction, colour-coded diagnostics that flag process problems (cold or over-oxidised heats, poor foaming, low basicity, energy-heavy heats) and recommend specific set-point changes, what-if and scenario evaluation, and safe operator-in-the-loop advice ahead of any closed-loop control. A working operator interface, diagnostics/guidance engine and sensitivity dashboard already exist as a starting point (Appendix B).

4.12 Industrial case studies and process-modification recommendations

Addressing O6 and the industrial mandate of the work, each phase culminates in a case study on the partner plant: benchmarking current practice, diagnosing losses with the calibrated model and sensitivity analysis, proposing modifications (charge mix, flux and lime practice, oxygen and carbon injection, power profile, foaming, DRI strategy, scheduling to cut power-off time), and — where trials are permitted — quantifying the realised improvement in kWh/t, yield, tap-to-tap time and emissions. Recommendations will be documented in a form directly usable by plant engineers and operators.

5 Proposed Joint Master's / Doctoral Programme

5.1 Structure and articulation

The programme is designed so that a Master's project delivers a complete, validated and industrially useful first platform, and a Doctoral project extends it into original, high-impact research. The Master's establishes the consolidated model, the calibration/sensitivity/validation methodology on one plant, and the operator decision-support tool with a first improvement case study. The Doctorate advances the hybrid physics-ML and soft-sensing science, real-time rigorous thermochemistry, DRI/H2-DRI extension and validation, integrated electrical-thermal-chemical modelling, and optimisation/advanced control with multi-plant validation and digital-twin deployment. The split is expressed through seven work packages (WP1–WP7).

5.2 Work packages

WP	Title	Objective and principal methods	Outputs	Phase
WP1	Model consolidation & software platform	Consolidate modular static+dynamic model; engineer control-ready, tested, documented platform with operator GUI (O1)	Reference model + platform; documentation	Master's
WP2	Data, calibration, sensitivity & UQ	Curate plant heat database; parameter estimation, Bayesian calibration, local+global sensitivity, UQ; cross-plant transfer test (O2)	Calibrated model; calibration & SA/UQ methodology; validation report	Master's → PhD
WP3	Validation & first industrial case study	Whole-heat and trajectory validation on held-out heats; diagnose losses; recommend & (if permitted) trial modifications (O6)	Validation; case study with quantified savings	Master's
WP4	Hybrid physics-ML & soft sensors	Soft sensors for T/C/FeO; thermodynamic surrogates; gray-box/PINN discrepancy models; data assimilation (O3)	Soft sensors; hybrid model; accuracy study	PhD
WP5	Real-time rigorous thermochemistry	Couple EERZ/zone-diffusion chemistry via fast surrogates; add O, S, CaS species (O1, O3)	Upgraded chemistry module; accuracy/runtime study	PhD
WP6	DRI / H2-DRI extension & validation	Extend & validate melting/refining for high-DRI, H2-DRI, Indian sponge iron; scenario studies (O4)	Validated DRI/H2-DRI model; practice recommendations	PhD
WP7	Optimisation, control & digital-twin deployment	Integrated electrical-thermal-chemical model; multi-objective optimisation; NMPC/RL; operator digital twin; industrial demonstration (O5, O6)	Control layer; digital twin; demonstration & impact report	PhD

5.3 Master's scope and milestones (≈ 18–24 months)

- M1 (0–4 mo): literature consolidation; platform set-up; partner data agreement and first data ingestion (WP1, WP2).
- M2 (4–10 mo): calibration and sensitivity/UQ on the partner plant; static+dynamic validation against held-out heats (WP2, WP3).
- M3 (10–16 mo): operator decision-support tool; first process-diagnosis case study and modification recommendations (WP3).
- M4 (16–24 mo): trial (if permitted) and quantification of improvement; dissertation and a conference/journal paper.

5.4 Doctoral scope and milestones (≈ 36–48 months)

- Y1: hybrid physics-ML soft sensors and data assimilation; deepen calibration/UQ and cross-plant transfer (WP4, WP2).
- Y2: real-time rigorous thermochemistry via surrogates; integrated electrical–thermal–chemical coupling (WP5, WP1).
- Y3: DRI/H2-DRI extension and multi-plant validation; scenario and green-steel studies (WP6).
- Y4: optimisation and NMPC/RL control; digital-twin deployment and industrial demonstration; thesis and journal papers (WP7).

5.5 Indicative timeline

Work package	M0–6	M6–12	Y2	Y3	Y4	Deliverable focus
WP1 Platform	●	●	○			Reference model & platform
WP2 Calib./SA/UQ	●	●	●	○		Calibrated model, methodology
WP3 Validation & case		●	●			Validation, savings case
WP4 Hybrid ML / soft sensors		○	●	●	○	Soft sensors, hybrid model
WP5 Thermochemistry			●	●		Fast rigorous chemistry
WP6 DRI / H2-DRI			○	●	●	Green-steel charges
WP7 Control & digital twin			○	●	●	Control, demonstration

● primary activity ○ supporting activity

5.6 Deliverables and success criteria

- **Deliverables:** a validated, calibratable static+dynamic EAF model and digital-twin/decision-support platform; a documented calibration/sensitivity/UQ methodology; validated hybrid soft sensors; a validated DRI/H2-DRI capability; an optimisation/control layer; industrial case studies; a Master's dissertation, a Doctoral thesis and peer-reviewed publications.
- **Success criteria:** agreed accuracy bands on tap temperature, carbon and specific energy on held-out heats; demonstrated cross-plant transfer; soft-sensor accuracy exceeding the physical model alone; and at least one industrially validated process modification with quantified energy/yield/productivity/emission benefit.

6 Expected Outcomes, Novelty and Impact

6.1 Scientific novelty

- A hybrid physics-informed machine-learning EAF model that couples a reduced-order dynamic model with soft sensors, thermodynamic surrogates and data assimilation — filling the most conspicuous gap in the current literature.
- A formal, reusable methodology for calibration, uncertainty quantification and cross-plant transfer of EAF process models.
- Real-time database-grade thermochemistry via surrogate models, and an integrated electrical-thermal-chemical formulation.
- Validated models and operating strategies for high-DRI and hydrogen-DRI charges, including Indian sponge-iron practice.
- Model-based optimisation and NMPC/RL control demonstrated on an industrial EAF.

6.2 Industrial relevance and benefits

The programme is explicitly designed to be useful to industry for studying their processes and recommending modifications. Expected benefits include reduced specific electrical and total energy (through better chemical-energy use, foaming, and reduced power-off time), improved metallic yield (through controlled bath oxidation and slag practice), higher productivity (shorter, more repeatable tap-to-tap times), lower electrode and refractory consumption, and reduced CO₂ and dust emissions — together with a transferable digital-twin and operator-guidance capability and a trained cohort of engineers fluent in EAF modelling and control.

6.3 Alignment with decarbonisation and Indian industry

By targeting DRI/hydrogen-DRI charges, energy efficiency and emissions, and by calibrating to local sponge-iron practice, the programme aligns with the global decarbonisation of steel and with the specific structure of the Indian electric-steel sector, where DRI-based EAF/induction routes dominate and energy and raw-material efficiency are decisive for competitiveness and sustainability.

6.4 Dissemination

Outcomes will be disseminated through peer-reviewed journal papers (process metallurgy, energy and control venues), conference presentations, the Master's dissertation and Doctoral thesis, and — subject to the partner's agreement and appropriate protection of intellectual property — through open methodology and software artifacts that lower the barrier for other plants to adopt model-based operation.

7 Resources, Collaboration and Risk Management

7.1 Required resources

- Industrial partnership providing historical and live heat data and, ideally, access for controlled trials and demonstration.
- Computing for CFD/thermodynamic reference runs and machine-learning training; standard workstation/cluster access.
- Thermodynamic database/software for the chemistry reference and surrogate training; laboratory access for validation analyses.
- Supervisory team spanning process metallurgy, thermochemistry, and modelling/control/data science; candidate(s) at Master's and Doctoral level.

7.2 Industry-academia collaboration model

A memorandum of understanding will define data sharing and governance, on-site access, trial protocols, publication and intellectual-property terms, and named technical liaisons at the plant. Regular review meetings will keep the research aligned with operational priorities and ensure that recommendations are practical and adopted.

7.3 Risk register

Risk	Likelihood	Impact	Mitigation
Insufficient or poor-quality plant data	Medium	High	Early data agreement; data-quality flagging; design-of-experiments; soft sensors robust to missing data
Model does not meet accuracy targets	Medium	High	Hybrid gray-box correction; targeted calibration on high-sensitivity parameters; staged acceptance criteria
Limited access for closed-loop trials	Medium	Medium	Operator-in-the-loop advisory deployment first; validate in simulator; demonstrate offline before online
Thermodynamic rigour too slow online	Medium	Medium	Fast surrogates; tiered fidelity (rigorous offline, surrogate online)
Cross-plant transfer weaker than hoped	Medium	Medium	Explicit transfer methodology; identify transferable vs plant-specific parameters; re-calibration workflow
Personnel / continuity across Master's→PhD	Low	Medium	Well-documented platform and data; overlapping scopes; reproducible software engineering

7.4 Ethics, intellectual property and data governance

Plant data will be handled under agreed confidentiality and governance terms; results intended for publication will be reviewed by the partner; and intellectual property arising will be managed under the institutional and partner agreements. No personal data is involved; the ethical considerations are principally around commercial confidentiality and the safe conduct of any plant trials.

8 Conclusion

The reviewed literature shows a mature and diverse EAF-modelling field, anchored by lumped-zone dynamic models that balance fidelity and speed, enriched by rigorous thermochemistry and deep CFD phenomenology, and increasingly concerned with DRI and hydrogen-DRI charges and with decision support. It also shows clear, field-level gaps: the fidelity/tractability trade-off, the scarcity of systematic calibration, uncertainty quantification and cross-plant transfer, the under-use of data-driven and hybrid methods, the disjointness of electrical and metallurgical models, the still-empirical treatment of foaming, and the distance from advisory to closed-loop control. The proposed joint Master's/Doctoral programme converts these gaps into a coherent research plan that is scientifically novel and directly useful to industry, and it can begin immediately on the software foundation already in place. The result will be both deep, transferable expertise in the EAF process and a validated, calibratable digital-twin and decision-support platform for studying and improving industrial EAF operations.

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Appendix A Structured Summary of the Reviewed Corpus

Categories: *R* = review; *S* = static/energy balance; *D* = dynamic lumped-zone; *T* = thermochemistry; *E* = electrical/arc; *C* = CFD/multiphysics; *DRI* = DRI/H₂-DRI process model; *DS* = decision support / control.

Work	Cat.	Approach & key contribution	Principal limitation
Logar et al. 2012, Pt 1 & 2	D/T	Zone energy balances; continuous heat/melt split; view-factor radiation; equilibrium+rate chemistry; validated	Empirical view-factor/kinetic coefficients; single-plant fit
MacRosty & Swartz 2005	D	Equilibrium zones with mass-transport limits; continuous, robust, optimisation-ready; parameters from plant data	Equilibrium-zone abstraction; needs parameter estimation
Bekker et al. 1999	D/DS	Multivariable non-linear state-space model for off-gas-based control	Chemically/thermally coarse (early model)
Opitz & Treffinger 2016	D/E	Whole-plant Modelica model (vessel, electric, electrode, off-gas); spatially discretised solid	Added complexity; simplified chemistry
Hay, Echterhof & Visuri 2019	D/DS	Comprehensive dynamic model + simulator; auto operation-chart generation; for training & scenario scans	Data-hungry; many empirical sub-models
Hay, Reimann & Echterhof 2019	T	Zone/diffusion-limited slag-metal chemistry; adds O, S, CaS; activity models; validated	Higher runtime; database dependence
Echterhof, Hay & Pfeifer (DynEAF)	D	Variable melting geometry; improved radiation & gas-phase; adaptable carbon sources	Development report; complexity
Van Ende 2022 (EERZ)	T	Effective Equilibrium Reaction Zone; kinetics + thermodynamic database; endpoint & hot-heel	Computationally heavy; reaction-zone parameters
Lozynskyy et al. (Energies)	E/DS	Real-time electrical-mode model; MPC of electrode movement; grid-impact analysis	Abstracts metallurgy
González et al. 2010	C/DRI	CFD + Lagrangian DRI-particle melting; frozen shell; feed/size/temperature effects	Expensive; particle-scale idealisation
Pineda-Martínez et al. 2015	C/DRI	Sponge-iron pellet melting in non-reactive slag; convective HTC controlling	Idealised (single pellet, non-reactive)
Chen et al. (CFD scrap)	C	3-D scrap-pile melting (dual-cell/stack) + arc + coherent jet; validated on 150-t EAF	Very high computational cost
Chen 2022 (PhD)	C	Integrated multi-physics EAF modelling	Not for online use
Morales et al. 2001	DRI	DRI melting simulator; cost-optimised balances → ODEs; carbon-fines bath-oxidation control	Route-dependent; single plant
Morales et al. 2004	DRI/T	FeO reduction by dissolved & injected carbon; surfactants; Dynamic Foaming Index	Empirical foaming/DFI
Li, Wang et al.	DRI	Mixed H ₂ -DRI + scrap process model; DRI share/chemistry effects on dynamics & yield	Needs broader DRI-quality validation
Decision-support system	DS	Model-based online monitoring & set-point advice	Single-plant; open-loop
AIST online balance	S/DS	Dynamic energy/material balance for on-line optimisation	Balance-level detail
Energy-efficiency analysis	S	Energy-stream decomposition & efficiency of the EAF	Static; fixed fractions
Conejo 2024 (monograph)	R/S	Comprehensive energy engineering & reduction methods	Reference, not a plant model
Hay et al. 2020 (review)	R	Review of mathematical EAF process models	Review
Makki Abadi et al. 2024 (review)	R	Review of simulation & numerical EAF modelling (CFD, energy, optimisation)	Review

Appendix B Existing Software Foundation

A working software platform, developed as a precursor to this proposal, already realises much of the WP1 scaffolding and demonstrates feasibility. It provides a starting point on which the programme can build from day one.

- **Static model** — a full per-heat mass and energy balance (charge, flux and lime sizing to target basicity, oxygen and carbon demand, slag chemistry, metallic yield, and an itemised energy breakdown with specific electrical energy).
- **Dynamic model** — a control-oriented, lumped-zone time simulation (solid scrap, liquid metal, slag, gas) with the continuous heating/melting split and mass re-averaging after Logar et al., an immersion-melting closure, refining chemistry (regime-switching decarburisation, Si/Mn/P oxidation, FeO formation and carbon reduction), CO post-combustion and a foaming index that modulates arc-transfer efficiency; endpoint and fixed-profile modes with time-series output and plots.
- **Documented, modifiable parameters** — an operating- and technical-parameter registry (charge, fluxes, oxygen, carbon, power, targets; efficiencies, heat losses, kinetics, furnace data) with units, valid ranges, range-checking and built-in help, so the model is transparent and calibratable.
- **Operator guidance / decision support** — a diagnostics engine that classifies each heat against targets and benchmarks and issues colour-coded, actionable recommendations (e.g. lime addition to reach a target basicity, oxygen/carbon adjustments to control FeO and foaming, actions for cold or over-blown heats).
- **Sensitivity-analysis suite** — one-factor and two-parameter sweeps, response-surface heatmaps, tornado rankings and trajectory overlays across both static and dynamic models, to be extended with global (Sobol/Morris) methods in WP2.
- **Operator-desk GUI** — a desktop dashboard with live editing of operating parameters, embedded heat plots, colour-coded guidance and on-demand sensitivity studies — a direct precursor to the digital-twin deployment of WP7.

The programme will calibrate and validate this platform on real plant data, upgrade its chemistry and melting closures, add hybrid soft sensors and control, extend it to DRI/H₂-DRI charges, and mature it into a deployed, validated digital twin — the concrete embodiment of the research outcomes.